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How I Added Persistent Semantic Memory to Claude Code in 15 Minutes

Mem0 + MCP + ChromaDB — plus what happens when your “helpful” hook spawns seven parallel Claude instances

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I asked Claude Code why I chose Inngest over raw cron jobs for Drippery. It had no idea. I'd made that call weeks earlier in a different session, written nothing down, and now it was gone.

Claude Code is great at writing code. **It's terrible at remembering what I told it yesterday.**

Every new session starts from scratch. CLAUDE.md helps, but it's a flat file with no search — when I need "what did I decide about auth for the public API?", I'm grepping through a markdown file like it's 2005.

I wanted something that **captures decisions automatically and lets me search them in natural language** across all my projects.

> *Here's how I built it in 15 minutes with zero infrastructure — and how the same setup nearly overheated my MacBook Air M4 a week later.*

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drippery.app

The Problem: AI Amnesia

I work with Claude Code daily. Over weeks, I've accumulated dozens of architectural decisions for Drippery:

- Why I chose Inngest over raw cron jobs
- Why the embed system is decoupled from sequences
- Why does the public API use x-api-key instead of OAuth

- Why I went with Resend over SendGrid

Every new Claude Code session, I either re-explain this or hope the CLAUDE.md catches it. **It usually doesn't.**

```
# Drippery — project notes for Claude Code

## Stack
- Next.js 16 App Router, TypeScript strict mode
- PostgreSQL via Drizzle ORM (migrations in `/db/migrations`)
- Clerk for auth (multi-tenant, organization-based)
- Stripe for billing, Resend for transactional email, Inngest for background jobs
- Render for hosting (Frankfurt region), Cloudflare R2 for image uploads
- Deployed via `main` branch auto-deploy, preview envs on PR

## Decisions

- Chose Inngest over raw cron jobs because I need retries, backoff, and step functions without running my own worker. Cron is cheaper but loses state on restart. 2
- Inngest function names must be `kebab-case`. The dashboard treats them as stable IDs, renaming breaks history.
- Resend over SendGrid — pricing is simpler ($0 for 3k/mo, $20 for 50k), API is cleaner, and the React Email integration saved me a template engine. SendGrid would
scale but I'm not there.
- Webhook signing for Resend: using `svix` library, secret in `RESEND_WEBHOOK_SECRET`. Do NOT parse webhook body before verifying signature — tried it, broke it, s
- Clerk organizations = tenants. Users can belong to multiple orgs. The active org is in the session — always read via `auth().orgId`, never from URL or form.
- Decided against custom roles in Clerk — using Clerk's built-in `admin` / `basic_member`. If we need more granularity later, switch to Clerk RBAC + our own permiss
- Drizzle schema: all tables have `id` (uuid v7), `createdAt`, `updatedAt`, `orgId` (FK to organizations). No soft deletes — use audit log table instead.
- UUID v7 chosen for time-sortable IDs. Makes pagination work without cursor shenanigans. 2026-02-03.
- Public API uses `x-api-key` header instead of OAuth. Reason: target user is a creator pasting a key into their Substack or blog, OAuth flow is overkill and scary.
revocable from dashboard. 2026-02-20.
- API keys are stored hashed (sha256), prefix is visible in UI for identification (`dk_live_abc1_`). Never log the full key.
- Rate limit: 100 req/min per key for free, 1000 req/min for paid. Redis-backed sliding window. Limits in `/lib/ratelimit.ts`.
- Embed widget is decoupled from sequences — it's a separate `embeds` table with a foreign key to a sequence. Reason: one sequence can be embedded in multiple plac
analytics. 2026-02-27.
- Embed widget ships as a single `<script>` tag that injects a shadow-DOM form. No iframe — iframes get blocked on Substack.
- Embed CSS uses only CSS variables + shadow DOM so it can't be styled by the host page. Tested on Substack, Ghost, WordPress, raw HTML.
- Confirmation emails are double opt-in by default, single opt-in is a Pro-only feature. GDPR needs this and I'm not fighting that fight.
- Unsubscribe link is REQUIRED in every send. Injected server-side, can't be overridden by template. If user tries to remove `{{unsubscribe_url}}`, the send API re
- Unsubscribe is one-click (CAN-SPAM compliant), doesn't require a confirmation page. Tracking pixel on open, link wrapper on click — standard email analytics.
- Bounce handling: Resend webhooks → Inngest → mark subscriber as `bounced` in DB. Hard bounces get suppressed globally. Soft bounces retry 3x over 72h.
- Complaints (marked as spam) → immediate suppression + flag the sequence for review. Three complaints in 24h = freeze sequence, notify owner.
- Decided to NOT build visual sequence editor. Sequences are defined as a list of steps with delay + template. Much simpler UI, fewer bugs. Revisit if users ask.
- Step delays are stored as ISO 8601 durations (`P1D`, `PT2H`). Inngest handles the scheduling.
- Minimum step delay is 5 minutes. Reason: protects against accidental spam loops. Max delay is 90 days.
- Max sequence length is 20 steps on Pro, 5 on Starter, 1 on Free. Hardcoded in `/lib/plans.ts`.
```

wall of text in CLAUDE.md

Why not Claude-Mem?

Before building my own, I tried claude-mem — the most popular persistent memory plugin for Claude Code with 46K+ GitHub stars. It's purpose-built, well-maintained, and does exactly what I need.

But I hit critical bugs on Apple Silicon. At the time I tested it, the MCP loopback wasn't saving observations reliably, the worker daemon kept timing out, and there were Bun-related crashes on macOS. Multiple open issues, no fix in sight.

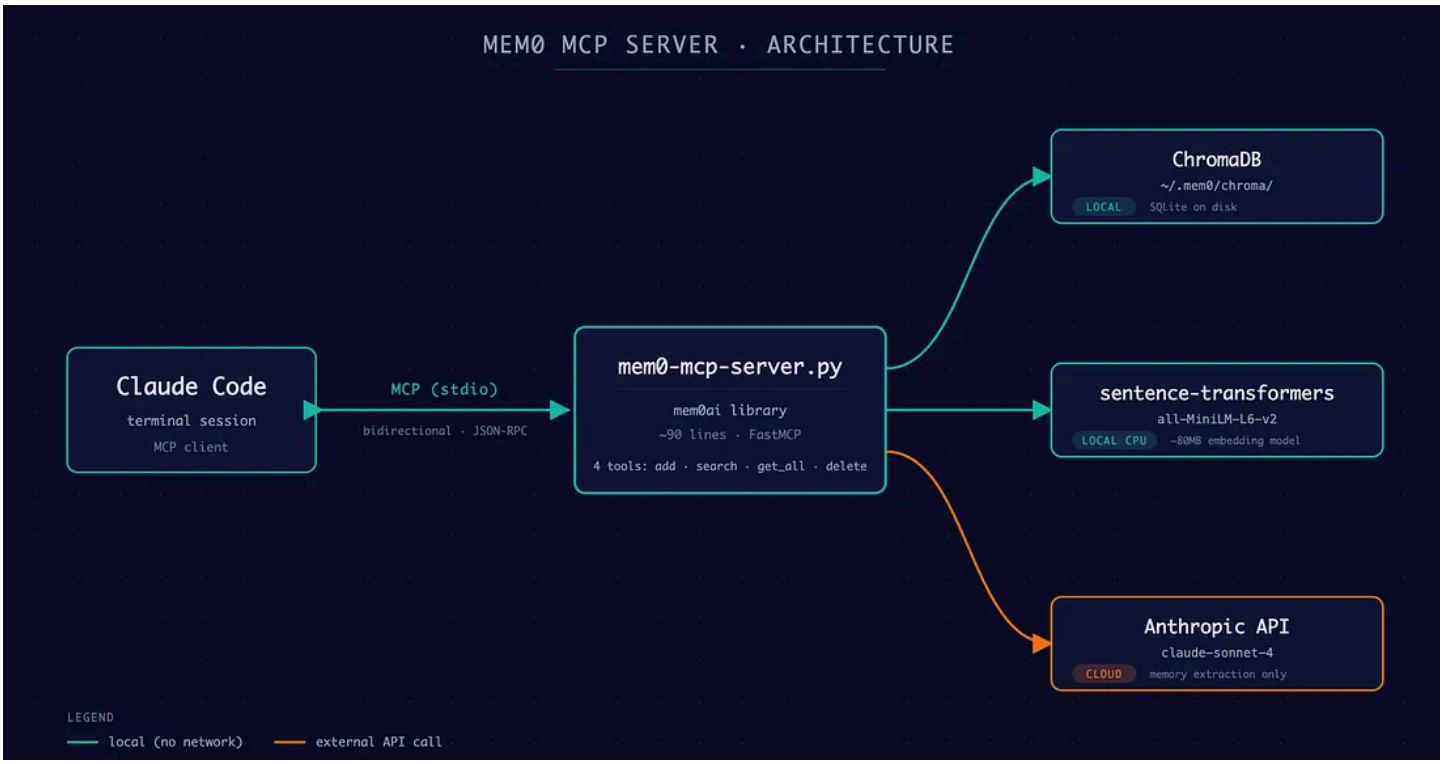
If you're on Intel Linux, it might work fine — on my M4 Mac, it was unusable. **So I built something simpler.** Fewer features, but it actually runs.

The Solution: Mem0 as an MCP Server

The fix is simple. **Three components:**

1. **Mem0** (mem0ai) — the memory layer. It takes text, extracts key facts using an LLM, stores embeddings in a vector database, and lets you semantic-search later.
2. **MCP (Model Context Protocol)** — the standard way to give Claude Code new tools. You write a server, and Claude gets new capabilities.
3. **ChromaDB** — local vector store. No Docker, no Qdrant, no PostgreSQL. Just a SQLite file on disk.

The architecture:



The architecture | Generated with Claude

Step 1: Install Dependencies

```
pip install mem0ai chromadb sentence-transformers mcp anthropic
```

That's it. **No Docker compose, no database setup, no config files.** sentence-transformers downloads the embedding model (~80MB) on the first run.

Step 2: Write the MCP Server

The entire server is ~90 lines of Python. Here's the core:

```python
#!/usr/bin/env python3
import os, sys, json
from mcp.server.fastmcp import FastMCP
CONFIG = {
    "llm": {
        "provider": "anthropic",
        "config": {
            "model": "claude-sonnet-4-20250514",
            "api_key": os.environ.get("ANTHROPIC_API_KEY", ""),
        },
    },
    "embedder": {
        "provider": "huggingface",
        "config": {"model": "all-MiniLM-L6-v2"},
    },
    "vector_store": {
        "provider": "chroma",
        "config": {
            "collection_name": "mem0",
            "path": os.path.expanduser("~/.mem0/chroma"),
        },
    },
}
_memory = None
def get_memory():
    global _memory
    if _memory is None:
        from mem0 import Memory
        _memory = Memory.from_config(CONFIG)
    return _memory

mcp = FastMCP("mem0")

@mcp.tool()
def mem0_add(text: str, user_id: str = "default", metadata: dict | None = None)
    """Store a memory — decisions, preferences, architecture choices."""
    result = get_memory().add(text, user_id=user_id, metadata=metadata or {})
    return json.dumps(result, default=str)

@mcp.tool()
def mem0_search(query: str, user_id: str = "default", limit: int = 5) -> str:
    """Semantic search across all stored memories."""
    results = get_memory().search(query, user_id=user_id, limit=limit)
    return json.dumps(results, default=str)
```

```python
@mcp.tool()
def mem0_get_all(user_id: str = "default") -> str:
    """List all stored memories for a user."""
    results = get_memory().get_all(user_id=user_id)
    return json.dumps(results, default=str)

@mcp.tool()
def mem0_delete(memory_id: str) -> str:
    """Delete a specific memory by its ID."""
    get_memory().delete(memory_id)
    return json.dumps({"deleted": memory_id})

if __name__ == "__main__":
    mcp.run(transport="stdio")
```

Save this as `~/.claude/mem0-mcp-server.py`.

Key design decisions:

- **Lazy loading** — the Memory object (and embedding model) loads on the first tool call, not on server start. This keeps the Claude Code startup fast.
- **HuggingFace embeddings** — all-MiniLM-L6-v2 runs locally, no API calls for embeddings. Only the memory extraction LLM hits Anthropic's API.
- **ChromaDB on disk** — data lives in `~/.mem0/chroma/`. Survives reboots, no server process needed.

Step 3: Register with Claude Code

```bash
claude mcp add mem0 -s user -- python3 ~/.claude/mem0-mcp-server.py
```

The `-s user` flag makes it available across all projects, not just the current one. **Restart Claude Code, and you'll see four new MCP tools.**

Step 4: Automate with a Hook

Manually calling "remember this" gets old fast. I wired a hook that runs a background `claude -p` instance on `SessionEnd` (and originally `PreCompact` too) — it reads the transcript, extracts non-obvious decisions, and stores them via `mem0_add`.

It worked. For about a week. Then my MacBook's fans ramped to max and wouldn't stop.

The Incident: Seven Parallel Claude Instances

One afternoon, my MacBook Air M4 — a machine that normally runs silently — **got uncomfortably hot.** The load average was 6.32 on a 10-core CPU.

I opened `top` and found the culprit:

```
claude -p "Extract key..."   72.9% CPU
claude -p "Extract key..."   66.1% CPU
claude -p "Extract key..."   44.9% CPU
claude -p "Extract key..."   36.8% CPU
claude -p "Extract key..."   25.0% CPU
claude -p "Extract key..."   19.1% CPU
claude -p "Extract key..."    —
```

Seven parallel headless Claude instances. Four `mem0-mcp-server.py` processes alongside them. All spawned by my own "helpful" memory hook.

```
daniel@macbook-air – top – 180×36

daniel@macbook-air ~ % top -o cpu -n 15
Processes: 647 total, 11 running, 636 sleeping, 3102 threads                    15:42:17
Load Avg: 6.32, 4.18, 2.75  CPU usage: 88.73% user, 11.27% sys, 0.00% idle
SharedLibs: 412M resident, 63M data, 30M linkedit.
MemRegions: 251837 total, 4287M resident, 178M private, 1523M shared.
PhysMem: 15G used (2857M wired, 1012M compressor), 221M unused.
VM: 172T vsize, 5214M framework vsize, 0(0) swapins, 0(0) swapouts.
Networks: packets: 1253147/1263M in, 798152/342M out.
Disks: 842512/19G read, 341247/8G written.

PID     COMMAND                %CPU   TIME      #TH  MEM     STATE       PPID
82341   claude -p "Extrahuj…"  72.9   00:04:12  47   984M    running     1
82156   claude -p "Extrahuj…"  66.1   00:03:58  44   912M    running     1
83102   claude -p "Extrahuj…"  44.9   00:02:14  39   856M    running     1
83245   claude -p "Extrahuj…"  36.8   00:01:49  37   821M    running     1
83891   claude -p "Extrahuj…"  25.0   00:00:58  34   754M    running     1
84012   claude -p "Extrahuj…"  19.1   00:00:32  32   712M    running     1
84187   claude -p "Extrahuj…"  –      00:00:02  15   124M    stuck       1
82350   python3 mem0-mcp-se…   12.4   00:01:23  8    342M    running     82341
82165   python3 mem0-mcp-se…   9.8    00:01:08  8    318M    running     82156
83110   python3 mem0-mcp-se…   6.2    00:00:47  7    286M    running     83102
83253   python3 mem0-mcp-se…   4.1    00:00:31  7    274M    running     83245
501     kernel_task            5.4    12:34:56  128  156M    running     0
234     WindowServer           2.1    03:21:45  18   284M    sleeping    1
1847    Electron Helper …      1.9    00:34:12  22   412M    sleeping    1842
1921    node                   1.2    00:12:08  9    186M    sleeping    1
```

Activity monitor | Generated with Claude

Root Cause: Four Bugs Stacked Together

I pulled the script apart and found **four issues:**

1. **Per-session lock, not global.** My concurrency guard was `/tmp/claude-mem0-locks/${SESSION_ID}.lock`. If five sessions ended at roughly the same time, each one had its own lock file — so each one happily spawned its own extraction. There was no global cap.

2. **PreCompact fires during sessions, not just at the end.** I had registered the same hook on both `SessionEnd` and `PreCompact`, assuming PreCompact was rare. It isn't. **It fires every time Claude Code compacts its context** — multiple times per long session.

3. **No hard timeout on the spawned process.** The `timeout: 10` in `settings.json` only applies to the shell wrapper. The wrapper returns instantly because it backgrounds the real work with `nohup ... &`. The actual `claude -p` instance? No limit. If it got stuck, it stayed stuck forever.

4. **No cleanup of orphaned MCP servers.** Each spawned `claude -p` started its own `mem0-mcp-server.py`. When the extraction hung, the Python server hung with it.

> *Any one of these on its own is survivable. All four together turn a memory hook into a runaway process.*

The Fix

I rewrote the hook with three changes:

1. Global concurrency cap via `pgrep`. Before spawning anything, check the entire process table:

```
RUNNING=$(pgrep -f "claude.*dangerously-skip-permissions.*Extrahuj klíčové" | w
if [ "${RUNNING:-0}" -ge 1 ]; then
    exit 0
fi
```

If any extraction is already running system-wide, skip. **One at a time, period.**

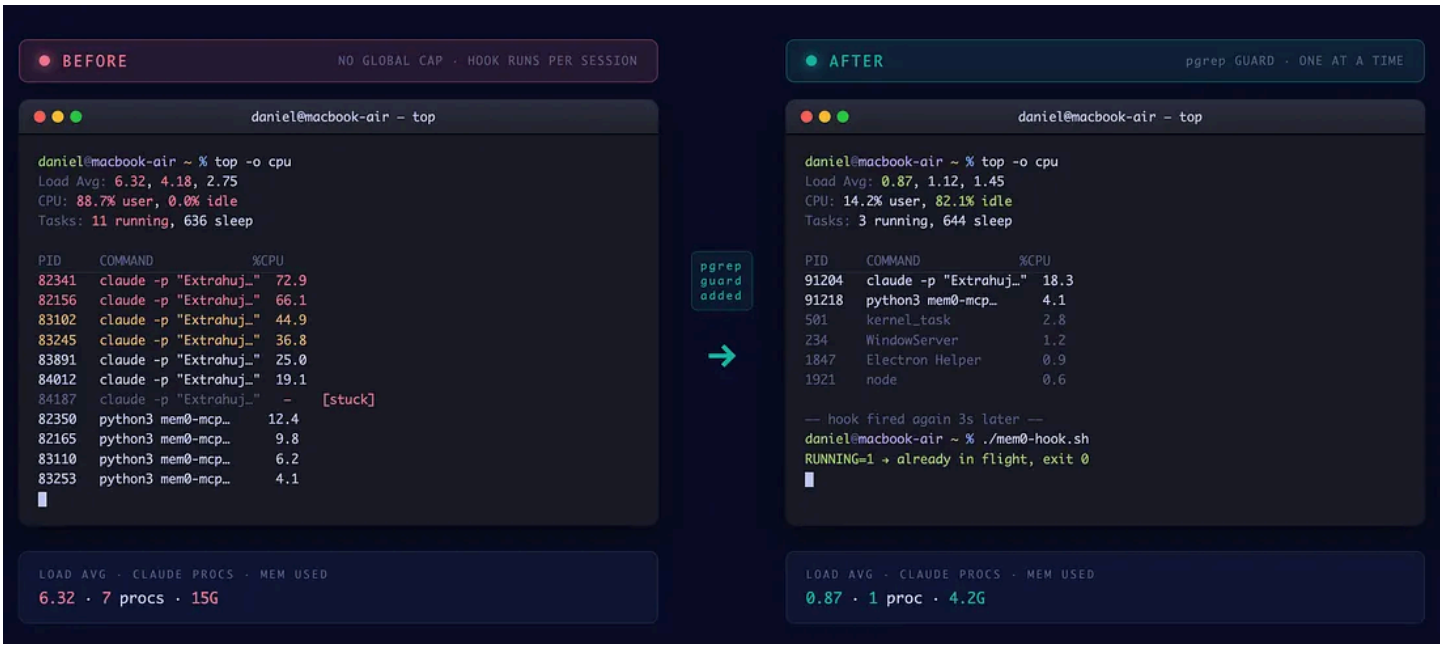
2. Remove the PreCompact hook entirely. `SessionEnd` alone is enough. If you want per-compact capture, you need your own throttle — don't just paste the same hook under both events.

3. Hard timeout on the spawned process. macOS doesn't ship `timeout` natively, so I detect `gtimeout` from coreutils:

```
TIMEOUT_BIN=""
for p in /usr/bin/timeout /opt/homebrew/bin/gtimeout /usr/local/bin/gtimeout; c
    if [ -x "$p" ]; then TIMEOUT_BIN="$p"; break; fi
done
nohup "$TIMEOUT_BIN" 300 "$CLAUDE_BIN" --dangerously-skip-permissions -p "$PROM
    > "$LOG_FILE" 2>&1 &
```

5-minute hard ceiling. If the extraction hangs, the OS kills it. Install with `brew install coreutils`.

After the fix, I re-tested by firing two hook calls back-to-back. Running instances after both fires: **1**. Global cap works.



Before/After Comparison | Generated by Claude

The Real Lesson

The original hook script wasn't written by me. **It was written by Claude Code, at my request, and I installed it without reading it carefully enough.** It worked on day one. It was broken by day eight — I just didn't know it.

The pattern is easy to miss: AI-generated infrastructure code looks correct, passes a dry-run, and **only fails under real-world concurrency that a single test can't reproduce.** These weren't syntax errors. They were wrong assumptions about the world:

- "Only one session ends at a time" — wrong, I often have 3+ Claude Code windows open.
- "PreCompact is rare" — wrong, it fires on every context compaction.

- “`timeout: 10` in settings.json contains the whole operation" — wrong, it only covers the wrapper.
- “nohup& is fire-and-forget and that’s fine” — wrong, fire-and-forget without cleanup is how runaway processes start.

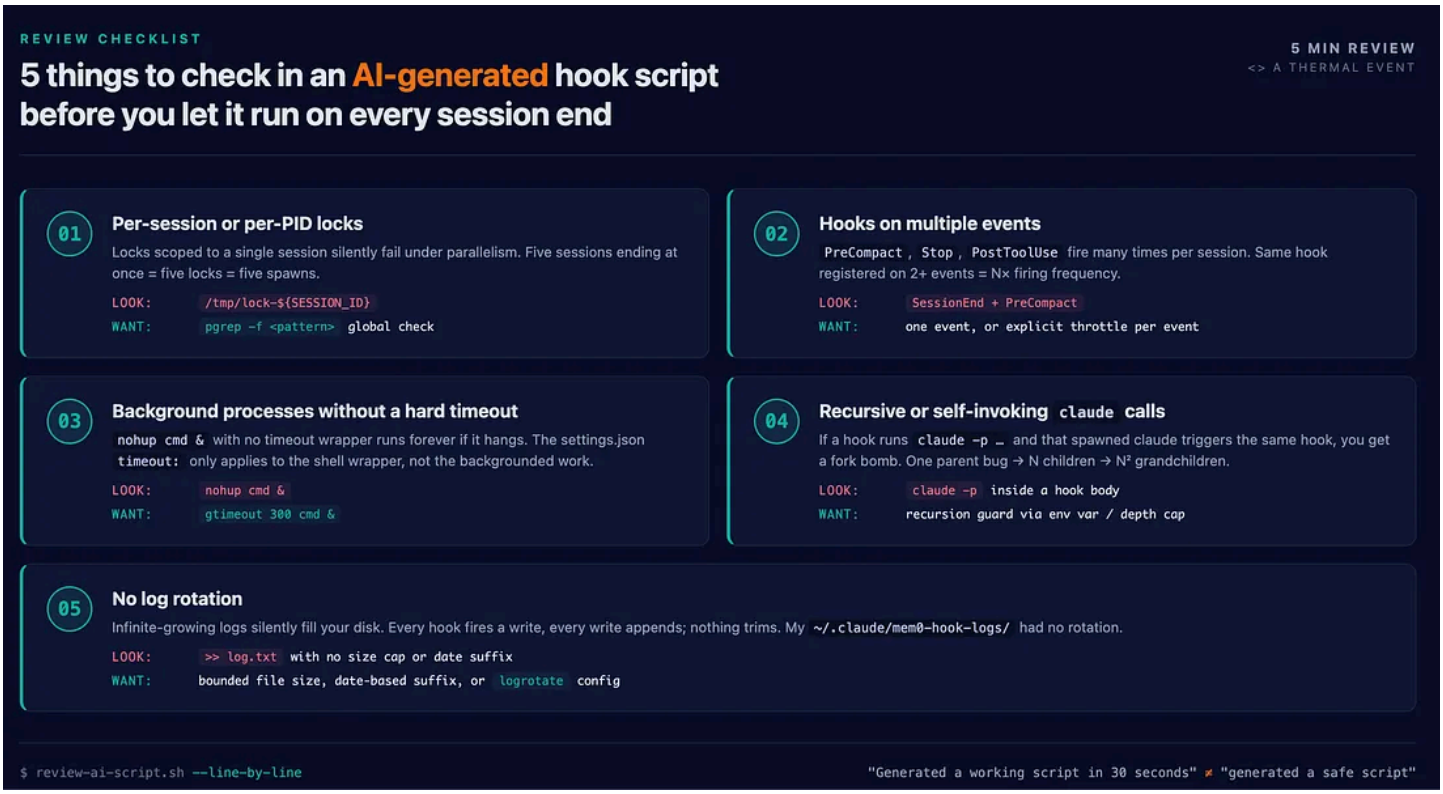
If you’re letting Claude Code install hooks, MCP servers, or anything that spawns background processes on your machine, **read the script line by line before running it.** Specifically look for:

- **Locks that are per-session or per-PID instead of global.** These silently fail under parallelism.
- **Hooks registered on multiple events** without thinking about firing frequency. `PreCompact`, `Stop`, `PostToolUse` — these can fire hundreds of times per session.
- **Background processes without a hard timeout.** `nohup cmd &` with no `timeout` wrapper is a liability.
- **Recursive or self-invoking `claude` calls** inside hooks. One bug in a parent session can spawn N children, each of which can spawn their own.
- **No limit on log files.** Mine were writing to `~/.claude/mem0-hook-logs/` with no rotation.

None of this is a reason not to use Claude Code, or not to let it write hooks. I still use it for exactly this. But **“it generated a working script in 30 seconds” is not the same as “it generated a safe script.”**

Review it like any PR — line by line.

My MacBook is fine. But I spent an afternoon diagnosing an overheating laptop that **a five-minute code review would have prevented.** Don’t be me.



Checklist graphic summarizing the 5 things to look for when reviewing AI-generated hook scripts | Generated by Claude

What This Gets You (When It's Working)

After a week of use, I can do things like:

- "What did I decide about auth?" → gets the x-api-key decision with context
- "Show me all architecture decisions for Drippery" → filtered by project
- "Why did I choose Inngest?" → semantic match, even though I never stored "Inngest" as a keyword

The data persists between sessions, between projects, and between machine reboots. It's just a SQLite file.

What It Doesn't Do

- **Not a replacement for CLAUDE.md** — CLAUDE.md is deterministic (always loaded), Mem0 is probabilistic (search-based). Keep both.
- **Costs tokens** — each mem0_add calls the Anthropic API for memory extraction. The extraction hook adds real cost per session end — budget for it.
- **Cold start** — first tool call loads the embedding model (~3 seconds on M1). Subsequent calls are instant.
- **No project isolation by default** — memories are global. Use the metadata field to tag by project if you need filtering.
- **Will happily overheat your laptop if you let it** — see above.

Try It — Carefully

The full setup takes 15 minutes:

1. `pip install mem0ai chromadb sentence-transformers mcp anthropic`
2. Save the MCP server script to `~/.claude/mem0-mcp-server.py`
3. `claude mcp add mem0 -s user -- python3 ~/.claude/mem0-mcp-server.py`
4. `brew install coreutils` (for `gtimeout`)
5. Write the hook script with a **global** `pgrep`-based concurrency cap and a `gtimeout 300` wrapper
6. Register it on `SessionEnd` only — not `PreCompact`
7. Restart Claude Code
8. **Read the script you pasted in, line by line, before you let it run on every session end**

Your AI coding assistant now has a memory. It's not perfect, but it's **much better than starting from zero every session** — and if you've wired up the guardrails, it won't spawn seven parallel processes behind your back.

Further Reading

- Mem0 — the memory layer used here (GitHub)
- Model Context Protocol — the open standard for Claude Code tools
- ChromaDB — local vector store, SQLite-based
- all-MiniLM-L6-v2 — the embedding model (Hugging Face)
- claude-mem — the alternative I tried first

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Andre Banki

What are your thoughts?